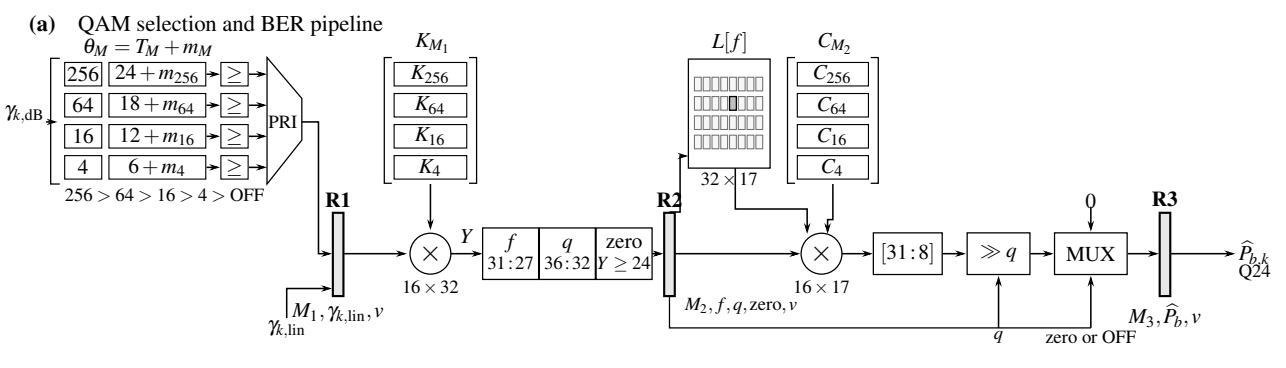

(a) QAM selection and BER pipeline
$\theta_M = T_M + m_M$
256
$24 + m_{256}$
64
$18 + m_{64}$
16
$12 + m_{16}$
4
$6 + m_4$
$\geq$
PRI
$\gamma_{k,\mathrm{dB}}$
256 > 64 > 16 > 4 > OFF
K_{M_1}
K_{256}
K_{64}
K_{16}
K_4
R1
$M_1, \gamma_{k,\mathrm{lin}}, v$
$\gamma_{k,\mathrm{lin}}$
$\times$
16×32
Y
f 31:27
q 36:32
zero $Y \geq 24$
R2
$L[f]$
32×17
C_{M_2}
C_{256}
C_{64}
C_{16}
C_4
$M_2, f, q, \mathrm{zero}, v$
16×17
[31:8]
$\gg q$
0
MUX
R3
$M_3, \widehat{P}_b, v$
q
zero or OFF
$\widehat{P}_{b,k}$
Q24